



SmartBursar

School Financial Management System.

Product Vision · 2026

THE APPLICATION

A complete **DIGITAL** school finance office.



Receive fees & issue receipts instantly



Track student balances in real time



Generate financial reports on demand



Maintain the cashbook automatically



Auditable, timestamped transaction log



Role-based access for administrators

Current Offline First Design



Runs entirely on one school PC



No internet required for any operation



No downtime, network outages don't matter



All data stays on school premises



Simple to install, no server setup



SmartBursar

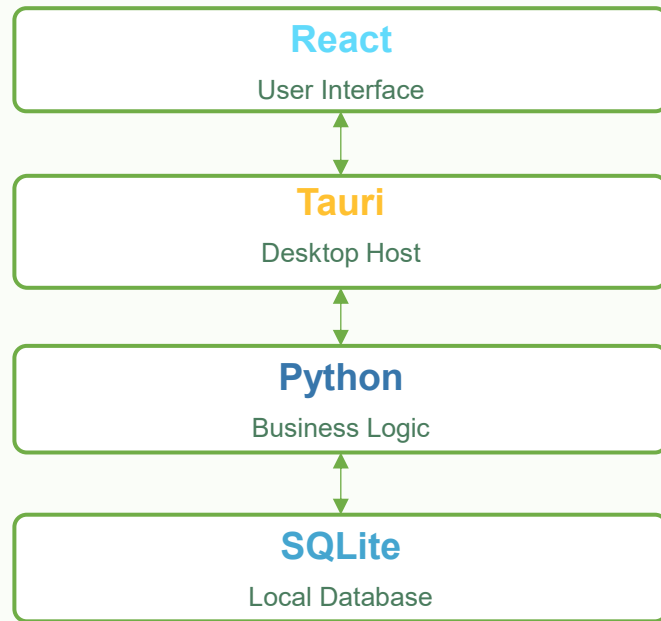
Desktop

The **pilot** product.

ONE school, **ONE** computer.

Current Phase.

- Single PC deployment
- SQLite local database
- Full offline operation
- Zero network configuration
- Receipts · Cashbook · Reports · Balances



SmartBursar

PLUS+

Multiple staff.

One **SHARED** financial system.

Across the **same** school network.

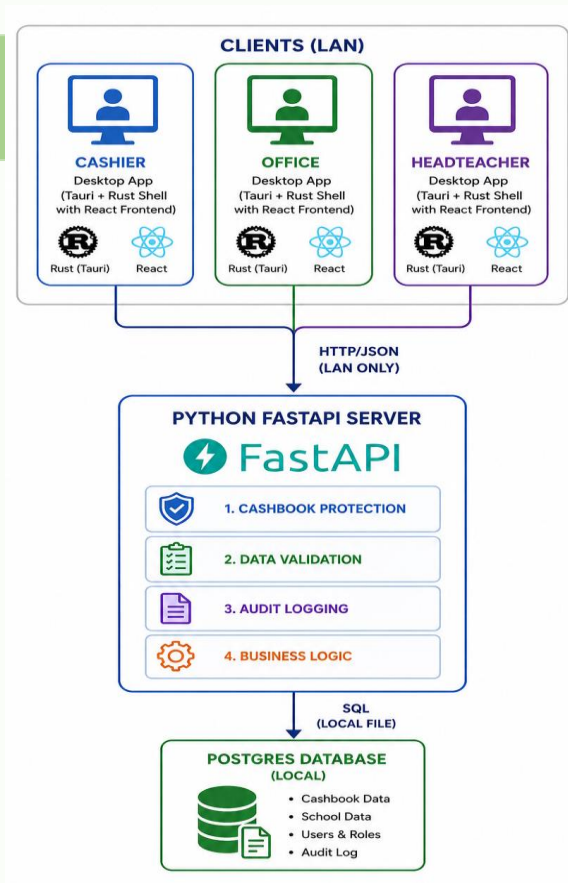
Multiple Concurrent users possible!

Bursar

Headteacher

Deputy

Director



SmartBursar

Enterprise

Multiple campuses.

Central oversight.

Cloud-based financial management.



Multi-campus

Multiple schools connected to a central system



Cross-school reporting

Aggregate analytics across all campuses



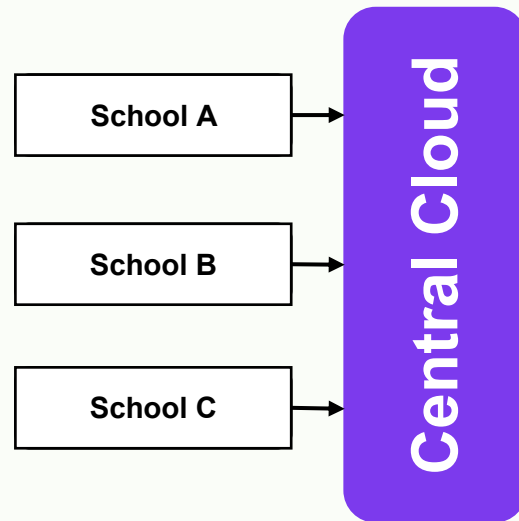
Cloud synchronization

Automatic backups and remote access



Proprietor access

Remote financial oversight from anywhere



TECHNOLOGY STACK

Purposefully chosen.

User Interface

React (Vite)

Desktop screens for bursar, cashier, headteacher, etc.

Desktop Host

Tauri + Rust

Packages the app as a cross-platform executable

Core Engine

Python

All financial logic lives here, independent of transport

Data Layer

SQLite / PostgreSQL

Secure local database, zero configuration for phase 1, and minimal adjustment for phase 2 and 3 to use PostgreSQL

Future Transport

HTTP / FastAPI / JSON

Optional adapter for phase 2 and 3 multi-user networking

Core Principle: The business logic doesn't know or care how it was called — IPC, HTTP, CLI. Everything else is just transportation.



ARCHITECTURE

Designed to **EVOLVE**,
Instead of be **rewritten**.

SmartBursar is built around a Hexagonal
(Ports & Adapters) architecture.

The Application Core, with all the business
rules, sits at the centre.

Transport is interchangeable.

RESULT:

Adding HTTP networking in Phase 2, or Cloud
Hosting in Phase 3 is an extension, not a
rewrite.

APPLICATION CORE

Payment Service

Student Service

Cashbook Service

Reports Service

SQLAlchemy ORM



